

Mapping the Consumption Gap

Planned Data Center Capacity versus Electricity Supply in California

Sebastian Oldman

M.S. capstone, Computer Science and Engineering, UC Santa Cruz

Defense deck. Data frozen September 22, 2026

The question, borrowed from Finland

- ▶ Finland (Manner, IRTF SUSTAIN, July 2026): about 285 MW of data centers operating, 7,777 MW in the public investment registry.
- ▶ California, December 2025: 23,277 MW of energization requests reported to the CEC, 20,677 MW in the state: 5,086 MW signed, 6,987 MW applications, 8,604 MW inquiries.
- ▶ Existing California data center load: about 1,000 MW, 7.0 to 9.3 TWh a year.
- ▶ The CEC discounts requests with confidence levels, a utilization factor and a ramp. Are the discounts right, can supply keep up, what can flexibility do, and how does the count compare with other regimes?

Sources: manner2026; cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; chapter 1 estimates (ch1_dc_load_estimates.csv).

Research questions and the method

- ▶ **RQ1** Pipeline by stage, absolute and relative to the grid.
- ▶ **RQ2** The 2030 gap, raw and probability-weighted, in TWh and net-peak MW.
- ▶ **RQ3** Six counting regimes compared; California's number under each rule.
- ▶ **RQ4** Headroom from load flexibility as a share of the gap.
- ▶ An independent replication and audit of the CEC method.
- ▶ Every raw file in a manifest (URL, access date, SHA-256); one command per chapter; every figure stamped with its files and manifest ids.
- ▶ Data frozen 2026-09-22; repository tagged; processed outputs exported with the report.

Chapter 1: the grid as it stands

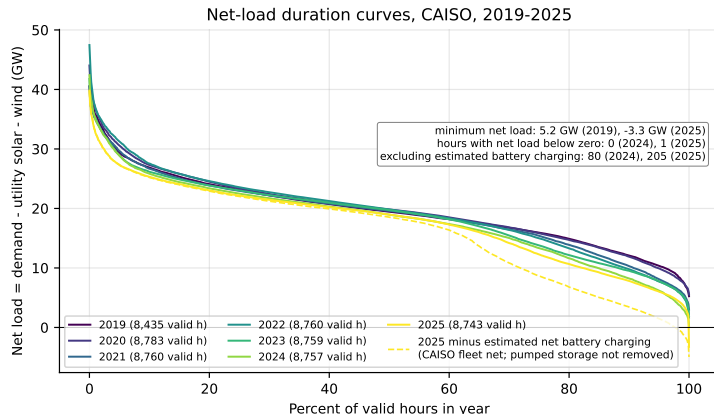


fig1_02. Data: ch1_net_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930 balance files; caiso_outlook fuelsource (Batteries). Code: ch1_baseline.add_storage_adjusted, duration_curves; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ CAISO demand 2019 to 2025 from EIA-930, cleaned against CAISO's own series; record peak 52,061 MW (2022), 43,860 MW in 2025.
- ▶ Net load after solar and wind now falls below zero for hundreds of hours a year; the evening net peak is what the fleet is planned for.

Chapter 1: prices, carbon and existing data center use

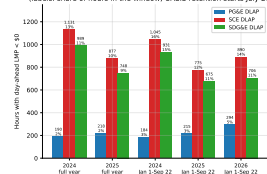
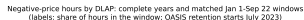
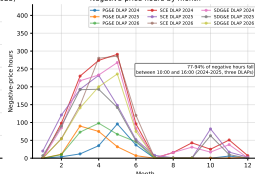


Fig 108. Data: ch0_negative_price_hours_windows.csv; ch1_negative_price_hours_by_month.csv; ch0_price_stats.csv. Sources: [manifest id]: [cable_news_data_large_monthly_files](#). Code: [ch1_negative_price_hours_by_month_price_stats_nut_chapter1_step2]. Data Ingest: 2023-08-22. [data/code/manifest.csv](#). [tools/IRI_access_data.py](#) and [IRI_2016_for_nut.py](#).



77-84% of negative hours fall between 18:00 and 16:00 (2004-2005, three DLAPs)

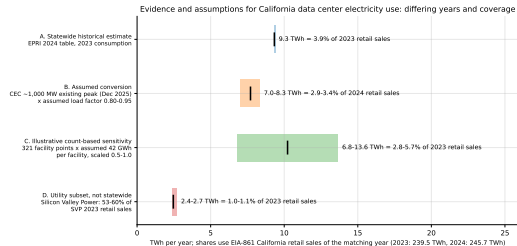


Fig1.88: Data: ch1_dk_lead_estimates.csv; ch1_c零售_sales_ea060.csv; ch1_5wp_fact_sheets.csv; ch1_kollar_grady_points_california_fig.csv; ch1_bottom_up_inputs.pptx. Sources: (modified) bto: april_powering_intelligence_2024; cdc; ch1_methodology_name_2020_ea060_2019_325w; 5wp_fact_sheets_2017_2020; kollar_grady_2025; smonod; epoch_data_centers. Code: ch1_baseline_dk_lead_estimates_bottom_up_estimates; c零售_sales_ea060_5wp_fact_sheets_eonr; kollar_grady_pont_chapter1; sheet_4. Data: From 2026-09-22: data/hub/manifest.csv; holds /IRL_ across_date and 914-756; data/hub/IRL_ across_date

- Negative day-ahead prices in 877 hours at the SCE load-aggregation point in 2025 (1,131 in 2024), 78 percent between 10 and 16 h.
- Existing data center use, four ways: a statewide estimate, an assumed conversion of the CEC's 1,000 MW, a count-based sensitivity and the Silicon Valley Power cluster. The spread is the finding.

Chapter 2: construction broke upward after ChatGPT

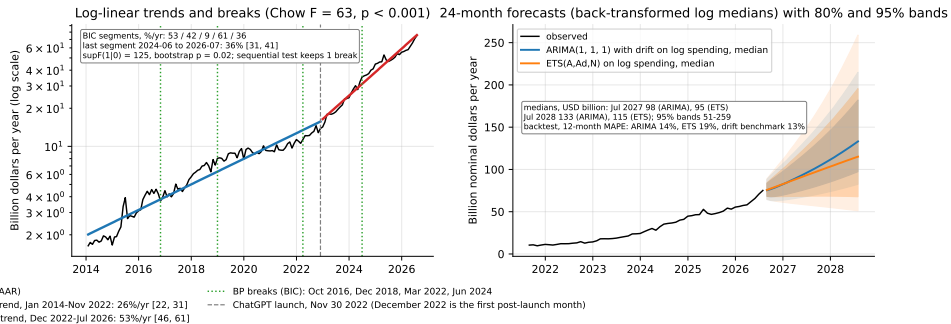


fig2_02 Data: ch2_census_break_summary.json; ch2_census_bai_perron.csv; ch2_census_segment_growth.csv; ch2_census_forecast_arima.csv; ch2_census_forecast_ets.csv; ch2_census_forecast_backtest.csv. Sources (manifest ids): census_c30_privsatime. Code: ch2_growth.chow_test, bai_perron, sequential_supf, arima_forecast, ets_forecast, backtest_forecasts; run_chapter2 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- Census data center construction spending: a Chow test at December 2022 and Bai-Perron breaks located by the data agree; ARIMA and ETS bands and a rolling-origin backtest.
- California has no state construction series; proxies are QCEW employment, CBRE Silicon Valley MW and the Epoch AI hub.

RQ1: the pipeline against the grid

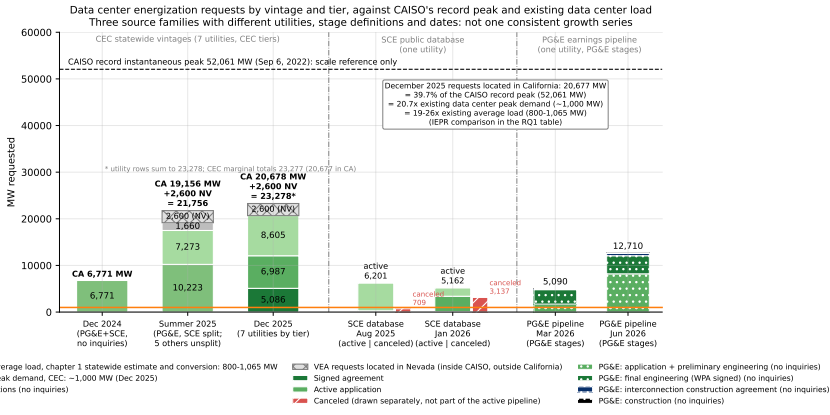
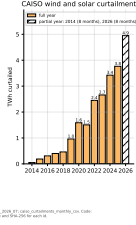
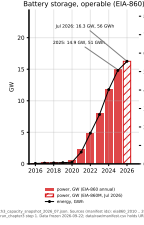
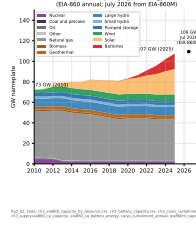
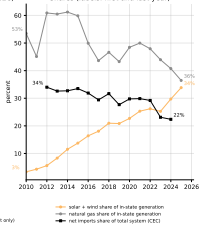
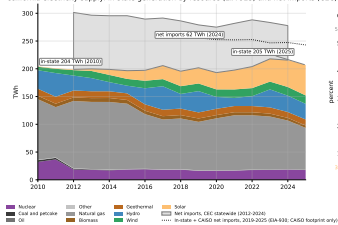


fig2_04. Data: ch2_tier_vintages.csv; ch2_rq1_denominators.json; ch1_dc_load_estimates.csv. Sources (manifest ids): cec_dc_forecast_2024iepr; cec_prelim_dc_forecast_2025; cec_dc_methodology_memo_2026; cec_assembly_hearing_2026_01_28; cec_tn266008; cec_tn268459; cec_tn272026; cec_tn272065; cec_tn272807; caiso_key_statistics_2026_08. Code: ch2_growth_tier_vintages; run_chapter2_steps_4_and_5. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ 20,677 MW in California: 40 percent of the CAISO record peak, twenty times the existing data center peak, five times the IEPR's 2025 to 2030 managed peak growth.
- ▶ VEA's 2,600 MW is in Nevada; SCE's cancellations grew from 709 to 3,137 MW between vintages.

Chapter 3: supply history and the fleet



- ▶ In-state generation 204 TWh (2010) and 205 TWh (2025); gas 53 to 36 percent, solar and wind 3 to 34 percent; imports 103 to 62 TWh (2012 to 2024).
- ▶ Capacity 73 to 107 GW; batteries 15 GW and 51 GWh; curtailment 3.8 TWh in 2025 and 4.9 TWh in January to August 2026.

Chapter 3: what gets built, from California's own record

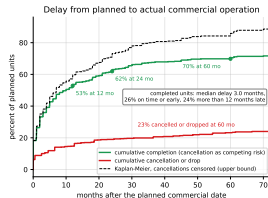
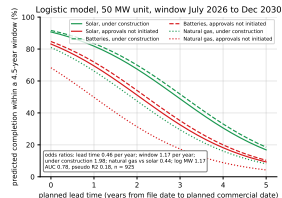
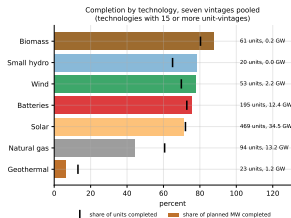
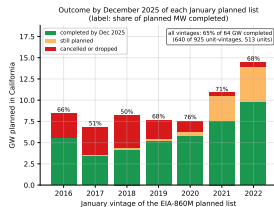


Fig 3.1. Data: ch3_vintage_outcomes.csv; ch3_realization_by_vintage.csv; ch3_realization_by_technology.csv; ch3_logit_coefficients.csv; ch3_logit_summaries.csv; ch3_delay.csv; ch3_delay_offset.csv; Sources: (transcript.txt), co2datahub, 2018, 03 ... 2022, 12; modelhub, 2018, 12 ... 2022, 12. Code: ch3_supply_and_demand_vintages_outcomes, 01_completion_logit_km_delay, net_chapter2_step3. Data frozen 2026-09-22; datahub/modelhub.csv holds vintages, access date and SHA-256 for each set.

- ▶ 925 unit-vintages of planned generators, January 2016 to 2022, followed to December 2025: 65 percent of megawatts operating; gas 44 percent, geothermal 6 percent.
- ▶ Logistic completion model (lead time, window, status, technology, size) and delay curves with cancellation as a competing risk.

Chapter 3: probability-weighted supply in 2030

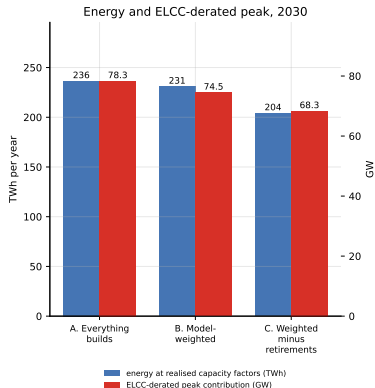
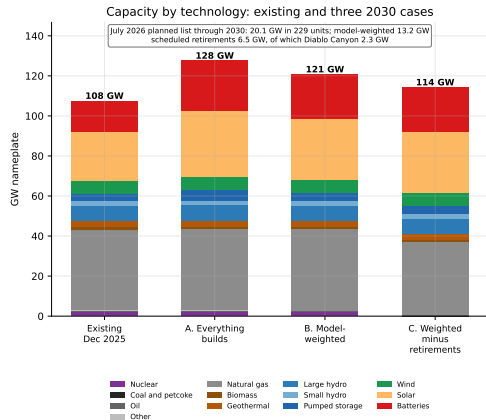


fig3_05. Data: ch3_cases_2030.csv; ch3_cases_2030_summary.json; ch3_planned_current_weighted.csv; ch3_existing_capacity_2025.csv; ch3_capacity_factors.csv; ch3_elcc_values.csv. Sources (manifest ids): els860m_2026_07; els860m_2025_12; cpuc_e3_astrape_incremental_elcc_2023. Code: ch3_supply/current_planned, cases_2030, capacity_factors, elcc_table; run_chapter3 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- Everything builds: 128 GW, 236 TWh, 78 GW of ELCC. Model-weighted: 121 GW, 231 TWh, 74 GW. Minus 6.5 GW of retirements including Diablo Canyon: 114 GW, 204 TWh, 68 GW.

Chapter 4: the same requests under four counting rules

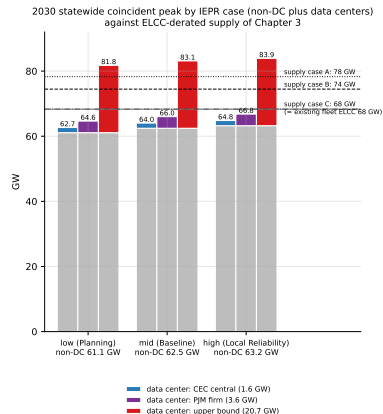
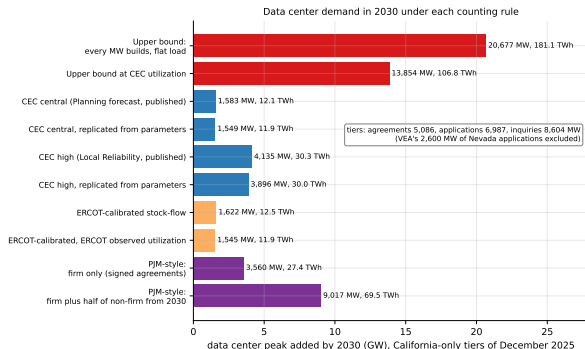


fig4_01. Data: ch4_demand_cases_2030.csv; ch4_demand_totals_2030.csv; ch4_cec_replication.csv; ch4_cec_ramp_profile.csv; ch4_ca_tiers.csv; ch4_iepr_growth_summary.csv; ch4_ercot_chain.csv; ch3_cases_2030_summary.json. Sources (manifest ids): cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; cec_in268722_268727 (CED 2025 forms); cec_in268124; cec_in268824; ERCOT decks and monthlies (see fig4_05); pjm_lar_summary_2025_11_24. Code: ch4_gap_ca_tiers, cec_parameters, cec_forecast_replication, ced2025_scenarios, iepr_growth_cases, ercot_chain, demand_cases_2030, demand_totals_2030; run_chapter4 steps 1-3. Data frozen 2026-09-22; data/rawmanifest.csv holds URLs, access date and SHA-256 for each id.

- Upper bound 20,677 MW and 181 TWh; CEC central 1,583 MW (rebuilt from its parameters: 1,549); CEC high 4,135; ERCOT-calibrated 1,622; PJM firm-only 3,560.
- The CEC parameters with SVP exempt reproduce its 2040 endpoints (4,855 and 7,381 MW) to within 1 MW.

RQ2: the 2030 gap, 10,000 draws

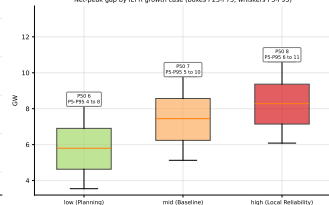
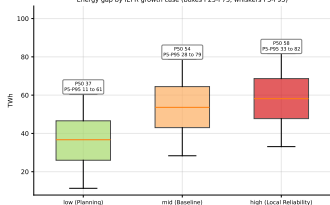
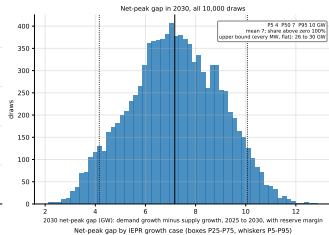
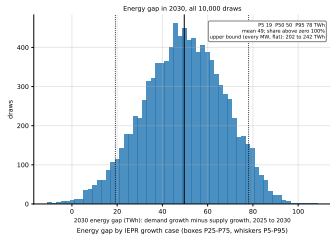
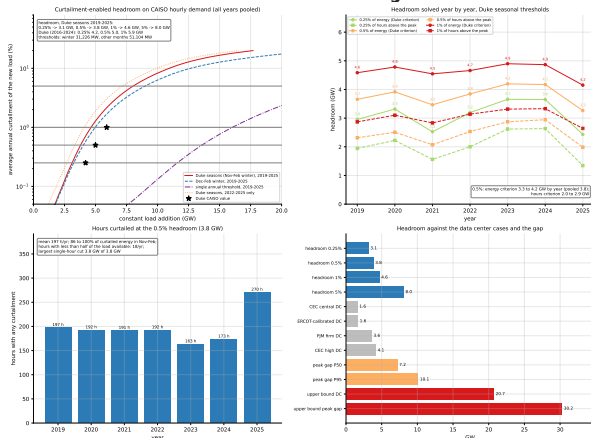


Fig4_S2. Data: ch4_mn, ch4n, parqut; ch4_mn_summary.csv; ch4_upper_bound_2035.csv; ch4_mn_inputs.csv; ch4_mn_key_numbers.json. Sources (manifest id): chapter 3 outputs: iia960m_2026_87, iia960m_2025_52, cpc, ei_sstnrc_incremental_elec_2023, iia923, cec, elec_energy_generation_page1; CBO 2025 forms as above. Code: ch4_cbo_SocioModel_mn_inputs, more, carls, evaluate, ssp, summaries, upper_bound_mn; run: chapter4_step4. Data from 2016-89-21: data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- ▶ Energy gap 50 TWh at the median (19 to 78); net-peak gap 7.2 GW (4.2 to 10.1). Data centers: 18 TWh and 2.3 GW of it.
- ▶ Upper bound: 202 to 242 TWh and 25.5 to 30.2 GW, four to five times the weighted 95th percentile. Gas at a 0.40 capacity factor would close the median energy gap.

RQ4: headroom from flexibility



Rep. Oil Data: rhd_headline_curve.csv; rhd_headline_summary.csv; rhd_headline_detail.csv; rhd_headline_by_year.csv; rhd_headline_by_gas.csv; rhd_energy_side.csv; rhd_yearly_2020_2025_clean_pumpout. Sources (marked in): statista, balance_2020 - 2025 (CRD adjusted demand), duke_refinishing, land_growth_2020, primes, method, CARGO values, cargo, east, dem, inc, (also outlinings monthly. (v. Data: rhd_gas_headline, headline by year, ton, chapter 400 8, data from 2020-09-01); datafromstatista.co/food/84/, access date and 104-156 for each id.

- ▶ Duke's method on CAISO load 2019 to 2025: 3.1, 3.8, 4.6 and 8.0 GW at 0.25, 0.5, 1 and 5 percent curtailment (Duke: 4.2, 5.0, 5.9); 98 percent of it in November to February.
- ▶ 3.8 GW is 2.4 times the CEC central data center addition, 53 percent of the median peak gap, 13 percent of the upper-bound gap.

Flexibility also cuts emissions

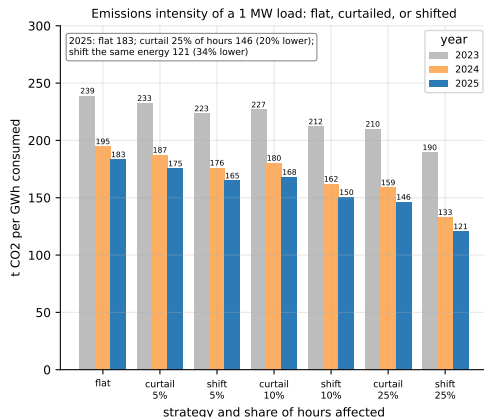
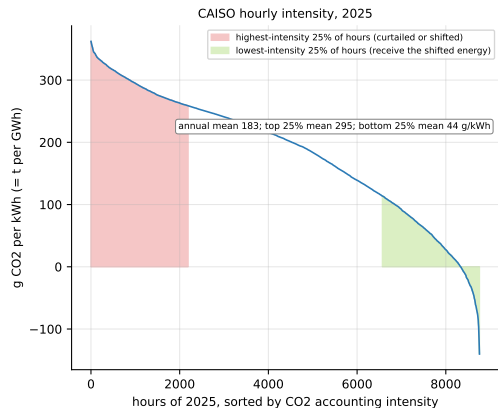


fig4_06. Data: ch4_emissions_flat_vs_flexible.csv; caiso_co2_intensity_hourly.parquet. Sources (manifest ids): caiso_outlook co2 and demand daily files. Code: ch4_gap.flat_vs_flexible_emissions; run_chapter4 step 8. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- A flat load carried 183 t of CO₂ per GWh in 2025; curtailing the highest-intensity quarter of hours cuts that by 20 percent, shifting the energy into the lowest-intensity quarter by 34 percent.

Contribution: a replication and audit, not a new forecast

- ▶ The CEC's published parameters rebuild its forecast: exact at the 2040 endpoints, within 2 to 6 percent in 2030 once the SVP exemption is applied.
- ▶ The supply side is discounted with the same discipline, from California's own completion record.
- ▶ The confidence levels are tested against ERCOT's realized transition rates and PJM's firm rule; the gap is a distribution with a stated upper bound and named drivers.
- ▶ Headroom replicated on CAISO load and expressed as a share of the gap; six regimes scored on one rubric.
- ▶ Everything reproducible: manifest, one command per chapter, stamped figures, frozen data, tagged repository, exported outputs.

Policy implications

- ▶ **CEC:** publish requests at project level with size, requested year and status history (as SCE does); report monthly by phase (as ERCOT does); publish firm and non-firm components separately (as PJM does); state the SVP exemption and the Nevada request where totals appear.
- ▶ **CPUC and CAISO:** a flexible interconnection service is worth 3 to 5 GW of headroom, mostly winter evenings, plus the midday surplus energy; FERC's June 2026 orders ask for exactly that.
- ▶ **Resource planning:** weight planned generation by completion probability, updated yearly from EIA-860M.
- ▶ **Legislature:** Diablo Canyon moves the gap more than any data center assumption (18 TWh, 2.2 GW); licences are in hand, state law stops at 2030.
- ▶ **EIA:** extend the pilot survey to California.

Limits

- ▶ Boundaries: CAISO hourly data against statewide generation and forecast; imports' peak contribution outside the absolute balance.
- ▶ ERCOT hazards measured during a surge; several ERCOT and PJM values read from chart images.
- ▶ ELCC from the CPUC's 2023 study; reserve margin an assumed range; Diablo Canyon a coin flip with the licence facts stated.
- ▶ Data center load flat at the peak; 2030 energy annualized at the end-2030 level.
- ▶ No project-level California request data outside SCE; the CEC's utility ramp schedules are not public.

Conclusion

- ▶ **RQ1** 20,677 MW in three tiers: 40 percent of the record peak, twenty times existing data center load.
- ▶ **RQ2** 2030 gap 50 TWh and 7.2 GW at the median (data centers 18 TWh, 2.3 GW); upper bound 202 to 242 TWh, 25 to 30 GW.
- ▶ **RQ3** Same requests: 20,677 / 5,086 / about 2,400 / about 1,000 MW under the four counting rules; regimes differ most in timeliness, verification and access.
- ▶ **RQ4** 3.8 GW of headroom at 0.5 percent curtailment: twice the weighted data center load, half the weighted peak gap, under a fifth of the upper bound.
- ▶ The method is the result: a forecast anyone can rebuild, a supply side discounted the same way, a gap with its uncertainty and drivers named.

Backup: reproducibility

- ▶ Repository: `src/` (`ch1_baseline`, `ch2_growth`, `ch3_supply`, `ch4_gap`, `ch4_regimes`, `provenance`), `scripts/run_chapter1..4.py`, `make ch1..ch4`, `make provenance`, `make freeze`, `make release`.
- ▶ Raw store: `data/raw/<group>/<access date>/`, immutable; `manifest.csv` with URL, access date, SHA-256; frozen copy `manifest_frozen_2026-09-22.csv`.
- ▶ Every figure stamped with its processed files, manifest ids and code; Appendix A repeats them as tables; `docs/DATA_NOTES.md` records every limit found.
- ▶ Standalone bundle on the Desktop reproduces every output byte for byte in its own environment.